

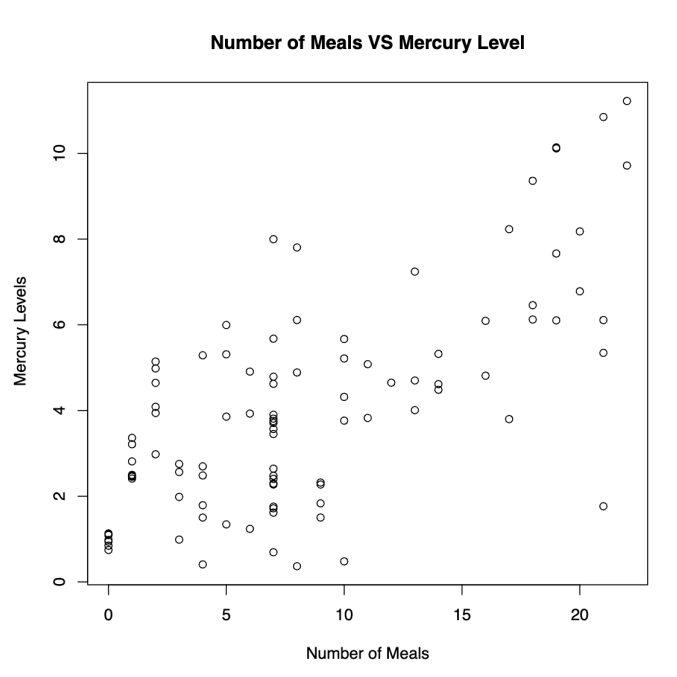
MET CS 555 Assignment 3 – 20 points

SUBMISSION REQUIREMENTS: **Please submit a single document (word or PDF) for submission.** Your submission should contain a summary of your results (and answers to questions asked on the homework) as well as your R code used to generate your results (please append to the end of your submission). Please use R for the calculations whenever possible. You will lose points if you are not utilizing R. You will also lose 10 points per day for late submissions unless prior arrangements are made with your facilitator.

The data in this document gives the number of meals eaten that contain fish (per week) and mercury levels in head hair for 100 fisherman. Save the data to a format that can be read into R. Read the data in for analysis. Use R to calculate the quantities and generate the visual summaries requested below.

(1) To get a sense of the data, generate a scatterplot (using an appropriate window, label the axes, and title the graph). Consciously decide which variable should be on the x-axis and which should be on the y-axis. Using the scatterplot, describe the form, direction, and strength of the association between the variables. **(4 points)**

The dots are scattered around the secondary diagonal. There seems to be a strong association between the two variables. The greater the number of meals, the stronger the association.

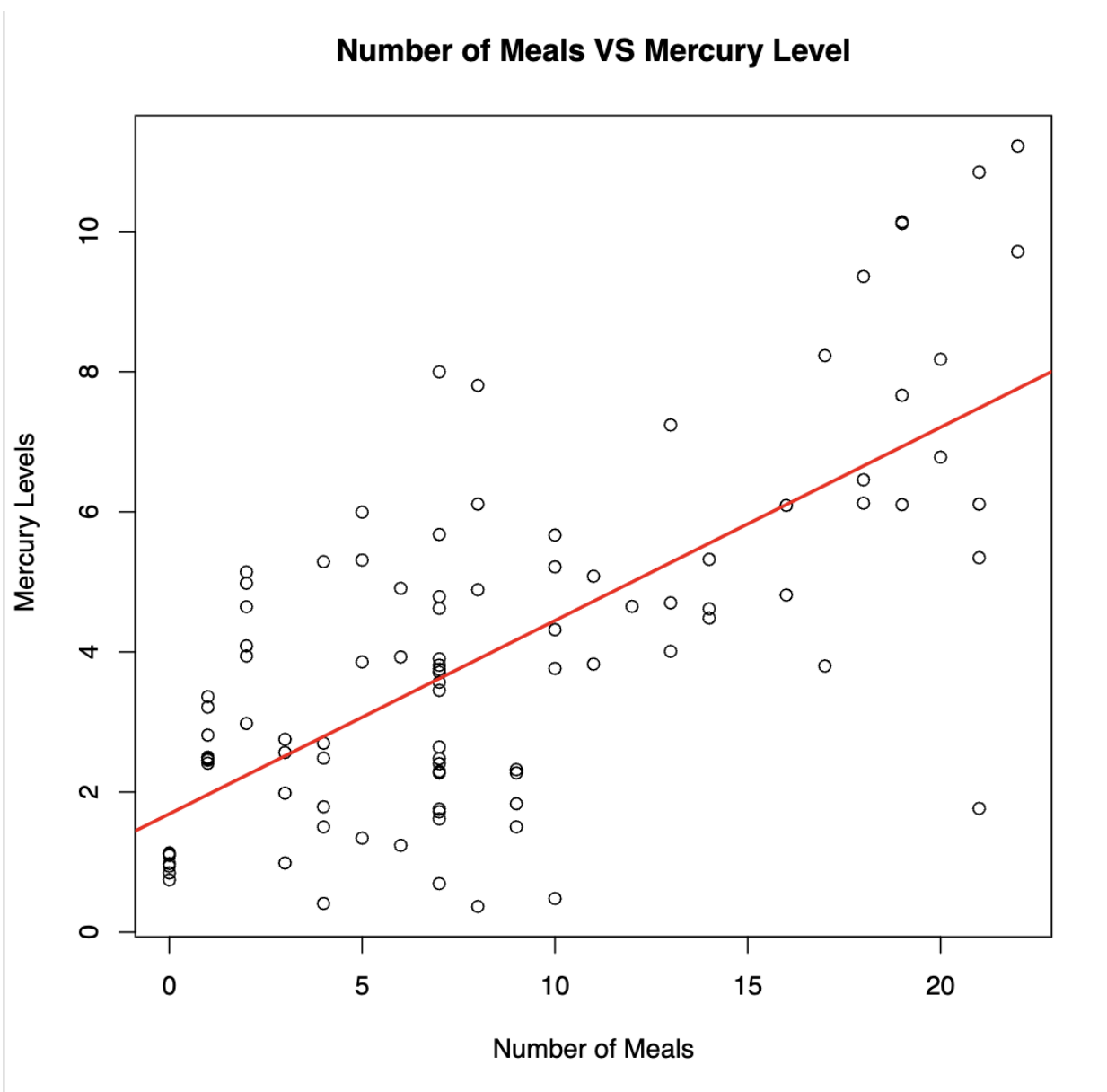


(2) Calculate the correlation coefficient. What does the correlation tell us? (2 points)

The correlation coefficient is 0.7. It tells us the linear association is strong - as the number of meals increases, the mercury level also increases.

(3) Find the equation of the least squares regression equation and write out the equation. Add the regression line to the scatterplot you generated above. (2 points)

$$y = 1.69 + 0.28x$$



(4) What is the estimate for β_1 ? How can we interpret this value? What is the estimate for β_0 ? What is the interpretation of this value? For the interpretations, you should be interpreting them in the context of this specific data set. (4 points)

Beta_1 = 0.28 means that the mercury level will increase by 0.28 for each meal.

Beta_2 = 1.69 is the average level of mercury in a person that doesn't eat fish at all.

(5) Calculate the ANOVA table **AND** the table which gives the standard error of $\hat{\beta}_1$. Formally test the hypothesis that $\beta_1 = 0$ using either the F-test or the t-test at the $\alpha = 0.05$ level. Either way, present your results using the 5-step procedure, as described in the course notes.

Within your conclusion, calculate the R-squared value and interpret this. Also, calculate (using R) and interpret the 90% confidence interval for β_1 . (8 points)

`anova(model)`

Analysis of Variance Table

Response: mercury_level

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
num_meals	1	309.24	309.239	93.689	6.013e-16 ***
Residuals	98	323.47	3.301		

`summary(model)`

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.68764	0.29833	5.657	1.53e-07 ***
num_meals	0.27595	0.02851	9.679	6.01e-16 ***

Hypothesis Testing

T Test

1

$H_0: \beta_1 = 0$

$H_1: \beta_1 \neq 0$

$\alpha = 0.05$

2 Select the following statistic

$$t = \frac{\hat{\beta}_1}{SE_{\hat{\beta}_1}}$$

with $df = 100 - 2 = 98$

3 Determine the critical value associated with a $df = 98$ and a right hand tail probability of 0.025

critical value = 1.98

if $|t\text{-statistic}| \geq 1.98$, reject the null hypothesis; else, fail to reject it.

4 $t\text{-statistic} = 9.7$

5 Since $|t\text{-statistic}| \geq 1.98$, we reject the null hypothesis. We have significant evidence at $\alpha = 0.05$ showing that there is a strong association between the number of meals and the mercury level.

Multiple R-squared: 0.4888. It indicates that 49% variability in mercury level is explained by the number of meals.

confidence interval at CI 95%

[0.2193744 0.3325262]

Data set for Assignment 3

Number of meals with fish	Total Mercury in mg/g
14	4.484
7	4.789
5	3.856
8	4.888
21	10.849
18	6.457
22	11.222
6	4.908
19	10.116

7	3.567
16	6.092
17	3.799
20	6.781
5	5.995
7	1.717
14	4.615
1	3.362
6	3.928
9	1.833
10	5.668
13	4.7
9	2.272
16	4.812
5	1.342
18	6.123
7	4.622
8	7.805
7	2.643
8	6.111
7	2.476
10	4.317
4	1.789
4	2.484
7	1.757
6	1.239
5	5.311
19	6.103
3	1.984
4	2.697
7	0.692
7	2.404
9	1.503
17	8.231
14	5.321
7	3.81
21	1.765
4	0.408
7	3.901
10	0.48
11	3.826
7	3.451
9	2.32

2	4.086
7	2.272
3	2.564
7	7.998
11	5.081
8	0.366
7	2.477
4	5.288
7	5.676
7	2.296
21	6.11
4	1.502
7	3.71
3	2.752
3	0.987
19	10.14
7	1.616
12	4.65
13	7.241
18	9.36
7	3.753
13	4.008
21	5.345
1	2.455
0	0.941
1	2.478
1	3.212
10	5.214
0	1.12
0	0.745
2	4.645
2	4.981
1	2.812
0	0.846
2	5.142
0	1.111
0	1.094
2	2.978
2	3.942
0	1.131
0	0.979
0	0.844
1	2.411

1	2.497
10	3.764
20	8.178
19	7.664
22	9.716

```
df <- read.csv("meals_mercury.csv", header = TRUE)

head(df)

degree_freedom = nrow(df)

cat(degree_freedom)

num_meals <- df$meals

mercury_level <- df$mercury

plot(num_meals,
      mercury_level,
      main = "Number of Meals VS Mercury Level",
      xlab = "Number of Meals",
      ylab = "Mercury Levels")
```

```
)  
  
cor(num_meals,mercury_level)  
  
model <- lm(mercury_level ~ num_meals)  
  
coef(model)  
  
abline(model,col = "red", lwd = 2)  
  
anova(model)  
  
cat("here")  
  
summary(model)  
  
# t value  
qt(0.975,98)  
  
# confidence interval  
confint(model,level = 0.95)
```